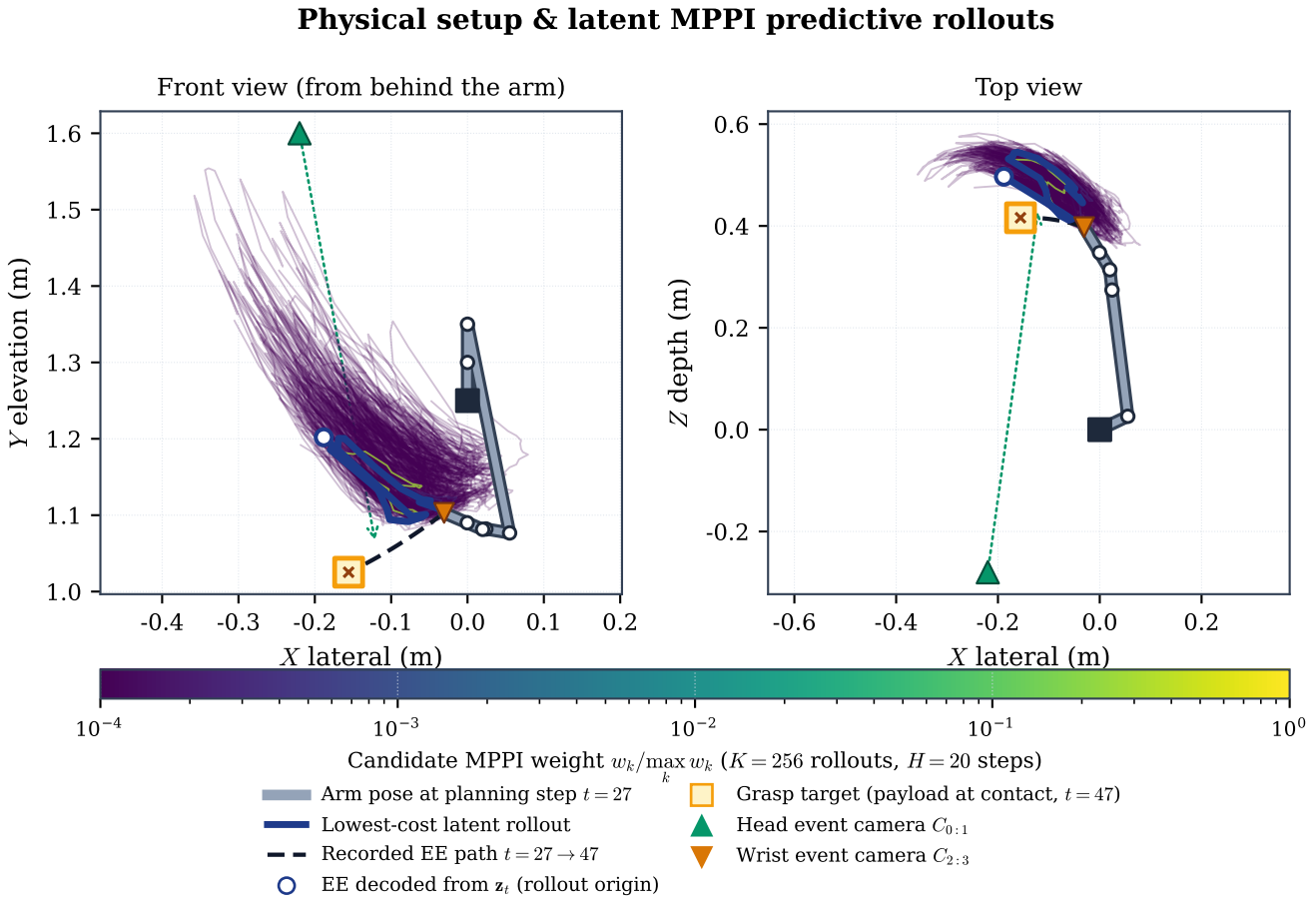

Physical setup & latent MPPI predictive rollouts
Front view (from behind the arm)
Top view
Y elevation (m)
Z depth (m)
X lateral (m)
X lateral (m)
1.6
1.5
1.4
1.3
1.2
1.1
1.0
0.6
0.4
0.2
0.0
-0.2
-0.4
-0.3
-0.2
-0.1
0.0
0.1
0.2
-0.6
-0.4
-0.2
0.0
0.2
10^{-4}
10^{-3}
10^{-2}
10^{-1}
10^{0}
Candidate MPPI weight $w_k / \max_k w_k$ ($K = 256$ rollouts, $H = 20$ steps)
Arm pose at planning step $t = 27$
Lowest-cost latent rollout
Recorded EE path $t = 27 \to 47$
EE decoded from $\mathbf{z}_t$ (rollout origin)
Grasp target (payload at contact, $t = 47$)
Head event camera $C_{0:1}$
Wrist event camera $C_{2:3}$